Cheat Sheet

Data Structures

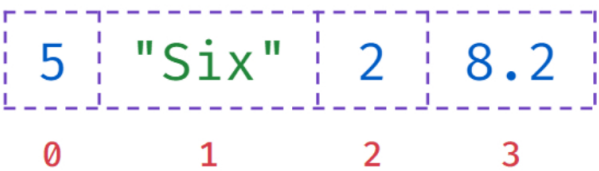
Data Structures allow us to store and organize data efficiently.
This will allow us to easily access and perform operations on the data.

In Python, there are four built-in data structures

- *List*
- *Tuple*
- *Set*
- *Dictionary*

List

List is the most versatile python data structure.
Holds an ordered sequence of items.



Creating a List

Created by enclosing elements within [square] brackets.
Each item is separated by a comma.

Code

PYTHON

```python
a = 2
list_a = [5, "Six", a, 8.2]
print(type(list_a))
print(list_a)
```

Output

```
<class 'list'>
[5, 'Six', 2, 8.2]
```

Creating a List of Lists

Code

PYTHON

```python
a = 2
list_a = [5, "Six", a, 8.2]
list_b = [1, list_a]
print(list_b)
```

Output

```
[1, [5, 'Six', 2, 8.2]]
```

Length of a List

Code

```python
a = 2
list_a = [5, "Six", a, 8.2]
print(len(list_a))
```

Output

```
4
```

Accessing List Items

To access elements of a list, we use Indexing.

Code

```python
a = 2
list_a = [5, "Six", a, 8.2]
print(list_a[1])
```

Output

```
Six
```

Iterating Over a List

Code

```python
a = 2
list_a = [5, "Six", a, 8.2]
for item in list_a:
    print(item)
```

Output

```
5
Six
2
8.2
```

List Concatenation

Similar to strings, `+` operator concatenates lists.

Code

```python
list_a = [1, 2, 3]
list_b = ["a", "b", "c"]
list_c = list_a + list_b
print(list_c)
```

Output

```
[1, 2, 3, 'a', 'b', 'c']
```

Adding Items to List

Code

```python
list_a = []
print(list_a)
for i in range(1,4):
    list_a += [i]
print(list_a)
```

Output

```
[]
[1, 2, 3]
```

Repetition

`*` Operator repeats lists.

Code

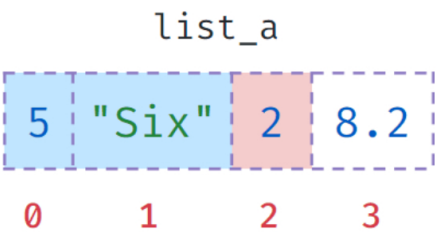
```python
list_a = [1, 2]
list_b = list_a * 3
print(list_b)
```

Output

```
[1, 2, 1, 2, 1, 2]
```

List Slicing

Obtaining a part of a list is called List Slicing.



Code

```python
list_a = [5, "Six", 2, 8.2]
list_b = list_a[:2]
print(list_b)
```

Output

```
[5, 'Six']
```

Extended Slicing

Similar to string extended slicing, we can extract alternate items using step

Similar to string extended slicing, we can extract alternate items using step.

Code

```python
list_a = ["R", "B", "G", "O", "W"]
list_b = list_a[0:5:3]
print(list_b)
```

Output

```
['R', 'O']
```

Converting to List

`list(sequence)` takes a sequence and converts it into list.

Code

```python
color = "Red"
list_a = list(color)
print(list_a)
```

Output

```
['R', 'e', 'd']
```

Code

```python
list_a = list(range(4))
print(list_a)
```

Output

```
[0, 1, 2, 3]
```

Lists are Mutable

- Lists can be modified.
- Items at any position can be updated.

Code

```python
list_a = [1, 2, 3, 5]
print(list_a)
list_a[3] = 4
print(list_a)
```

Output

```
[1, 2, 3, 5]
[1, 2, 3, 4]
```

Strings are Immutable

Strings are Immutable (Can't be modified).

Code

PYTHON

```python
message = "sea you soon"
message[2] = "e"
print(message)
```

Output

```
TypeError: 'str' object does not support item assignment
```

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